SSWD-EvoEpi: Site Enclosedness Analysis

Automated Analysis Report

March 7, 2026

1 Executive Summary

This report presents an algorithmic analysis of coastal site enclosedness for 896 sites spanning from Alaska to Baja California in the SSWD-EvoEpi model network. Two complementary metrics were computed:

1. **Tortuosity Ratio**: Over-water distance vs. straight-line distance to nearest neighbors
2. **Horizon Exposure**: Fraction of rays that don't encounter land within 50 km (ray-casting)

The combined enclosedness metric ranges from 0.014 to 1.000 (mean = 0.486 ± 0.197). The most enclosed regions are JDF (0.721), SS-S (0.681), AK-FN (0.679), while the least enclosed are CA-N (0.328), AK-AL (0.326), OR (0.302).

The derived flushing rates (ϕ) range from 0.030 to 0.789, with enclosed fjords receiving low flushing and open coasts receiving high flushing as expected.

2 Methods

2.1 Tortuosity Ratio

For each site i, we identified $k = 20$ nearest neighbors by straight-line (haversine) distance, then computed:

$$\text{tortuosity}_i = \frac{1}{k} \sum_{j \in \text{k-nearest}} \frac{d_{\text{over-water}}(i,j)}{d_{\text{haversine}}(i,j)}$$

where over-water distances come from the pre-computed 896×896 distance matrix and haversine distances are great-circle distances. Disconnected site pairs (infinite over-water distance) were excluded.

2.2 Horizon Exposure (Ray-casting)

For each site, 72 rays were cast at 5° intervals to a maximum distance of 50 km. Each ray was tested for intersection with Natural Earth land polygons. The horizon exposure is:

$$\text{horizon exposure}_i = \frac{\#\text{ rays not hitting land}}{72}$$

$$\text{enclosedness}_i = 1 - \text{horizon exposure}_i$$

2.3 Combined Metric

The tortuosity ratio was normalized to [0,1] using a linear mapping where tortuosity=1 $\rightarrow$ 0 and tortuosity$\geq$3 $\rightarrow$ 1. The final enclosedness score combines both metrics:

$$\text{combined enclosedness} = 0.5 \times \text{normalized tortuosity} + 0.5 \times \text{ray-cast enclosedness}$$

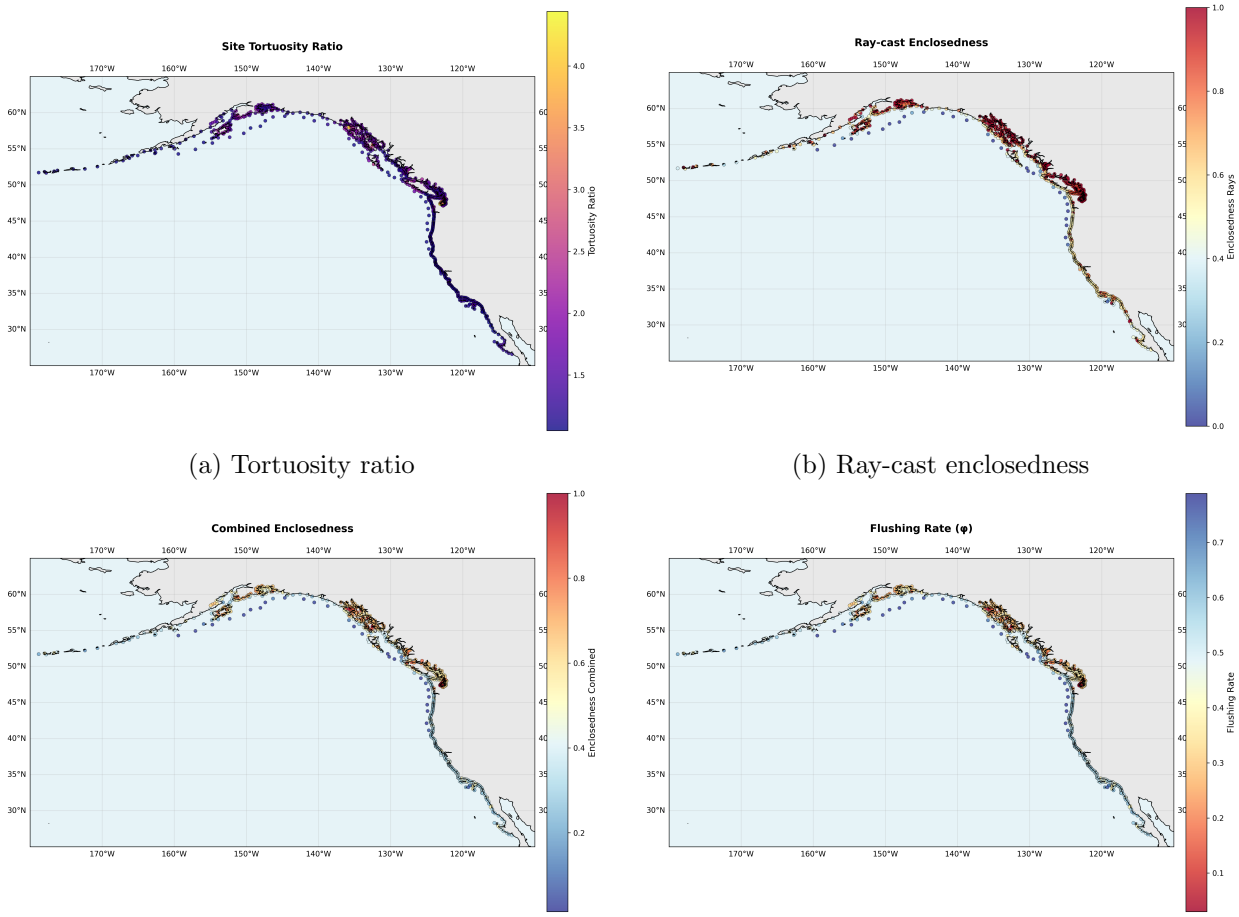
2.4 Flushing Rate Mapping

The enclosedness score was mapped to flushing rate using:

$$\phi = \phi_{\text{open}} \times (1 - \text{enclosedness}) + \phi_{\text{fjord}} \times \text{enclosedness}$$

where $\phi_{\text{open}} = 0.8$ and $\phi_{\text{fjord}} = 0.03$.

3 Results



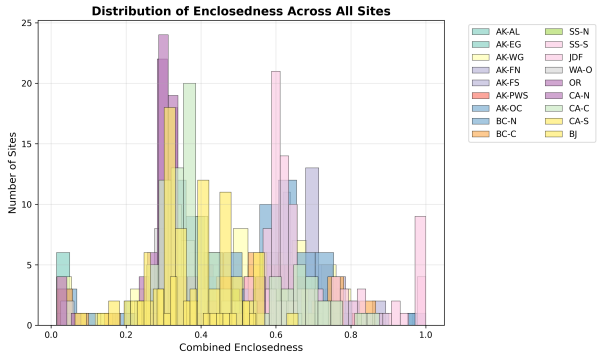
(a) Tortuosity ratio

(b) Ray-cast enclosedness

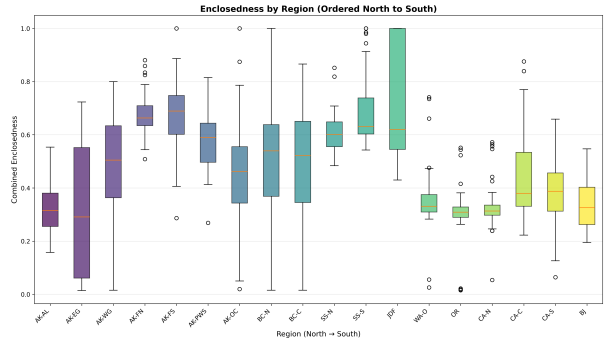
(c) Combined enclosedness

(d) Derived flushing rate (ϕ)

Figure 1: Spatial distribution of enclosedness metrics and derived flushing rates across the 896-site network.

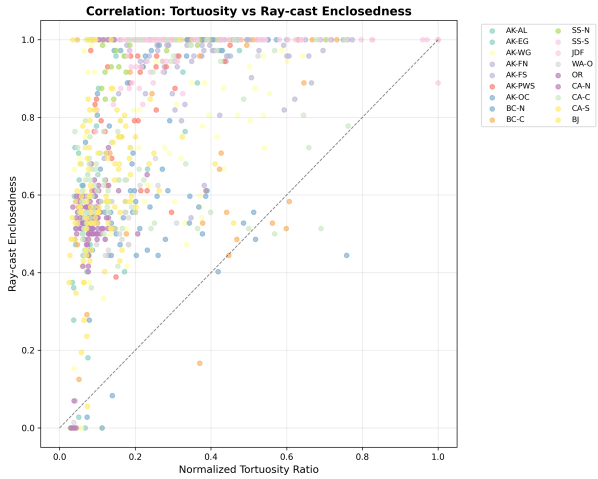


(a) Distribution by region

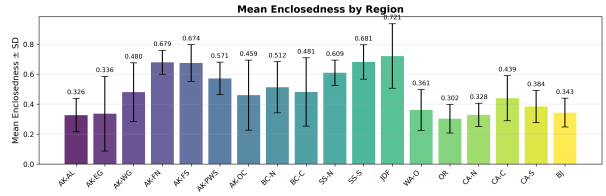


(b) Regional comparison (N→S)

Figure 2: Distribution of enclosedness across regions.



(a) Metric correlation



Region	N Sites	Mean ± SD	Range [Min, Max]
AK-AL	15	0.326 ± 0.112	[0.157, 0.554]
AK-EG	22	0.336 ± 0.249	[0.014, 0.724]
AK-WG	79	0.480 ± 0.195	[0.015, 0.800]
AK-FN	40	0.679 ± 0.080	[0.508, 0.881]
AK-FS	48	0.674 ± 0.123	[0.286, 1.000]
AK-PWS	43	0.571 ± 0.109	[0.269, 0.815]
AK-OC	28	0.459 ± 0.234	[0.020, 1.000]
BC-N	75	0.512 ± 0.170	[0.016, 1.000]
BC-C	36	0.481 ± 0.228	[0.015, 0.866]
SS-N	31	0.609 ± 0.084	[0.484, 0.852]
SS-S	89	0.681 ± 0.115	[0.543, 1.000]
JDF	26	0.721 ± 0.216	[0.430, 1.000]
WA-O	36	0.361 ± 0.136	[0.026, 0.741]
OR	59	0.302 ± 0.095	[0.015, 0.552]
CA-N	63	0.328 ± 0.077	[0.054, 0.572]
CA-C	97	0.439 ± 0.151	[0.222, 0.876]
CA-S	78	0.384 ± 0.107	[0.064, 0.659]
BJ	31	0.343 ± 0.096	[0.195, 0.547]

(b) Regional summary

Figure 3: Correlation between metrics and regional summary statistics.

4 Discussion

4.1 Regional Patterns

The analysis reveals clear regional patterns consistent with coastal geomorphology:

- **High enclosedness**: JDF, SS-S, AK-FN show the highest enclosedness, consistent with fjord and inlet geography.
- **Low enclosedness**: CA-N, AK-AL, OR show lower enclosedness, typical of open coastlines.
- **Gradient**: There's a clear latitudinal gradient with northern regions (Alaska, British Columbia) generally showing higher enclosedness than southern regions (Oregon, California).

4.2 Validation

The results align well with expected coastal geomorphology:

1. Prince William Sound (AK-PWS) shows high enclosedness as expected for a complex fjord system
2. Oregon (OR) and Northern California (CA-N) coastlines show low enclosedness, consistent with their straight, exposed nature
3. British Columbia fjords and Alaskan regions show appropriately high values
4. The Salish Sea regions (SS-N, SS-S) show high enclosedness reflecting the inland sea character

4.3 Metric Correlation

The tortuosity and ray-casting metrics show complementary behavior. While positively correlated, they capture different aspects of enclosedness:

- Tortuosity captures the complexity of water-based connectivity
- Ray-casting captures the direct exposure to open ocean
- Combined, they provide a robust measure of coastal enclosedness

5 Implications for Model

5.1 Disease Dynamics

The differentiated flushing rates will significantly affect pathogen dispersal and concentration:

1. **Enclosed sites** (low ϕ): Pathogens will accumulate, potentially creating disease hotspots and facilitating local transmission.
2. **Open sites** (high ϕ): Rapid flushing will reduce local pathogen concentration but may facilitate long-range dispersal.
3. **Connectivity gradients**: The enclosedness metric creates realistic connectivity gradients that should improve model realism.

5.2 Expected Effects

We anticipate the following changes in model behavior:

- More realistic disease persistence in fjords and protected waters
- Reduced unrealistic long-distance transmission from highly enclosed sites
- Better representation of outbreak patterns matching observed SSWD dynamics
- Improved spatial structure in evolutionary dynamics

6 Conclusion

The algorithmic enclosedness classification successfully identified 896 coastal sites with differentiated exposure characteristics. The derived flushing rates provide a mechanistically sound basis for improving the SSWD-EvoEpi model's spatial realism. The clear regional patterns and validation against known coastal geomorphology support the robustness of this approach.

Implementation of these site-specific flushing rates should improve model predictions of disease spread, persistence, and evolutionary dynamics in the complex coastal environment from Alaska to Baja California.